

Ideation Phase

Empathy Map Canvas

Date **19 June 2026**

Team ID -----

Project Name AI-Enhanced Intrusion Detection System

Maximum Marks 4 Marks

Empathy Map Canvas

An empathy map helps understand the needs, pain points, and expectations of users who interact with the intrusion detection system.

For this project, the main users are:

- **Network Administrators**
- **Security Analysts**
- **Organizations/Companies**

The goal is to understand their challenges in monitoring and detecting cyber threats.

Empathy Map for AI-Enhanced Intrusion Detection System

Think & Feel

- "Is my network secure?"
 - Concerned about unknown cyberattacks
 - Wants faster detection of malicious traffic
 - Needs reliable security alerts
 - Worried about data breaches
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Hear

- Security experts recommend stronger monitoring
- Organizations discuss increasing cyber threats

- Team members report suspicious activities
 - Clients demand better security protection
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See

- Large amount of network traffic daily
 - Growing number of cyberattacks
 - Traditional systems missing new attack patterns
 - Complex security logs
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Say & Do

- Monitors traffic regularly
 - Checks alerts and logs
 - Responds to suspicious activities
 - Updates firewall/security settings
 - Investigates attack reports
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Pain

- Hard to detect attacks manually
 - Too many false alarms
 - Slow response time
 - Complex log analysis
 - Risk of missing important threats
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Gain

- Real-time intrusion detection
- Faster attack identification
- Improved network security
- Reduced manual monitoring

- Better accuracy with AI/ML model
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Problem Statement

Traditional intrusion detection systems are slow and may fail to detect modern cyber threats accurately. Security teams need an intelligent solution that can analyze network traffic in real time and classify attacks effectively.

Proposed Solution

The **AI-Enhanced Intrusion Detection System** uses Machine Learning (Random Forest Classifier) to analyze network traffic data and detect attack types such as:

- Normal Traffic
- DoS Attack
- Probe Attack
- R2L Attack
- U2R Attack

The system provides:

- Live dashboard monitoring
- Real-time alerts
- Prediction logs
- API integration for external systems

